

GRIDGUARDIAN AI

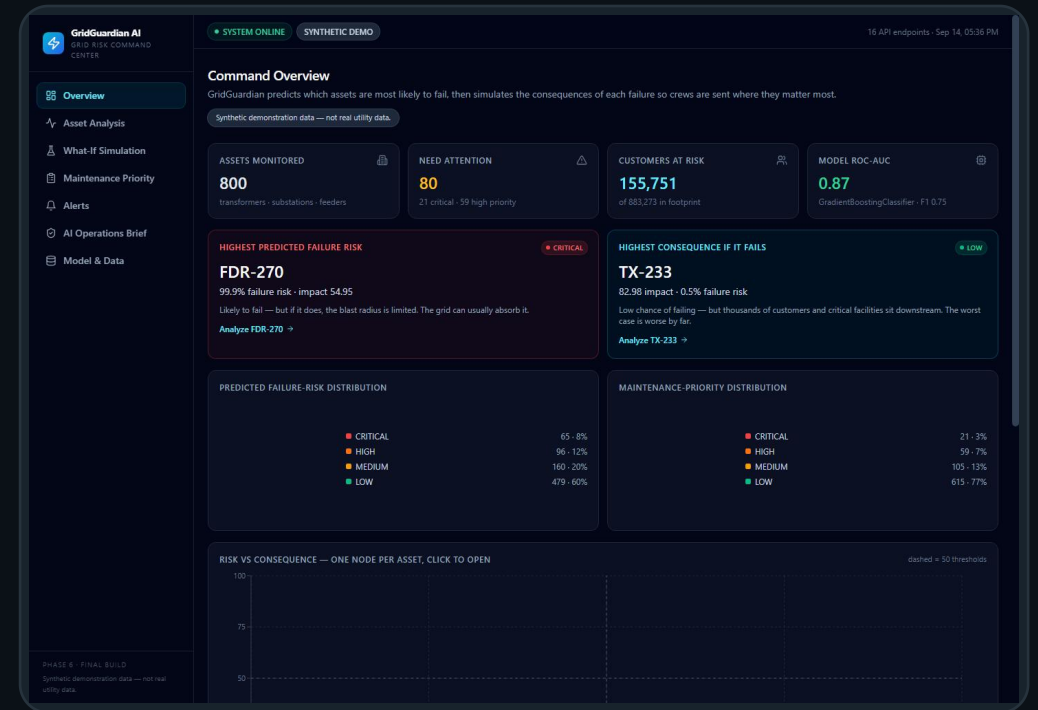
Predict. Simulate. Prioritize. Respond.

A decision copilot for grid operators — predict failure risk, simulate the cascade, then tell the crew exactly what to fix first.

TEAM VOLTVISION

Aum Vaja — Team Lead

Vrunda Kotadiya Tarang Prajapati Riza Bhatt



Live product view — GridGuardian operations dashboard

Operators know what might fail. Not what to do about it.

Utility networks stream sensor data for every asset — load, temperature, vibration, age. The backlog dwarfs the crew:

WHEN DOES IT HAPPEN?

Predicting *when* a unit fails is only the first step.

WHAT BREAKS WHEN IT FAILS?

A transformer can take down zones, hospitals, whole suburbs.

WHAT DO WE FIX FIRST?

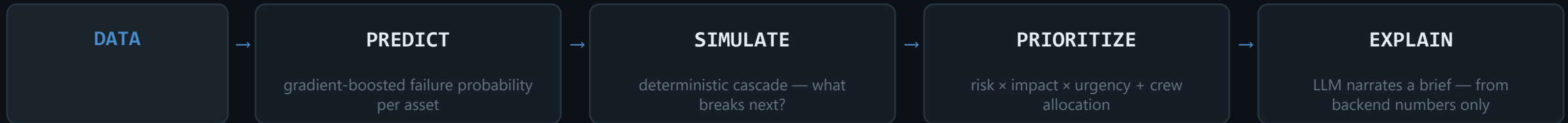
Three crews. 800 assets. The work order comes last.

THE DANGEROUS DEFAULT

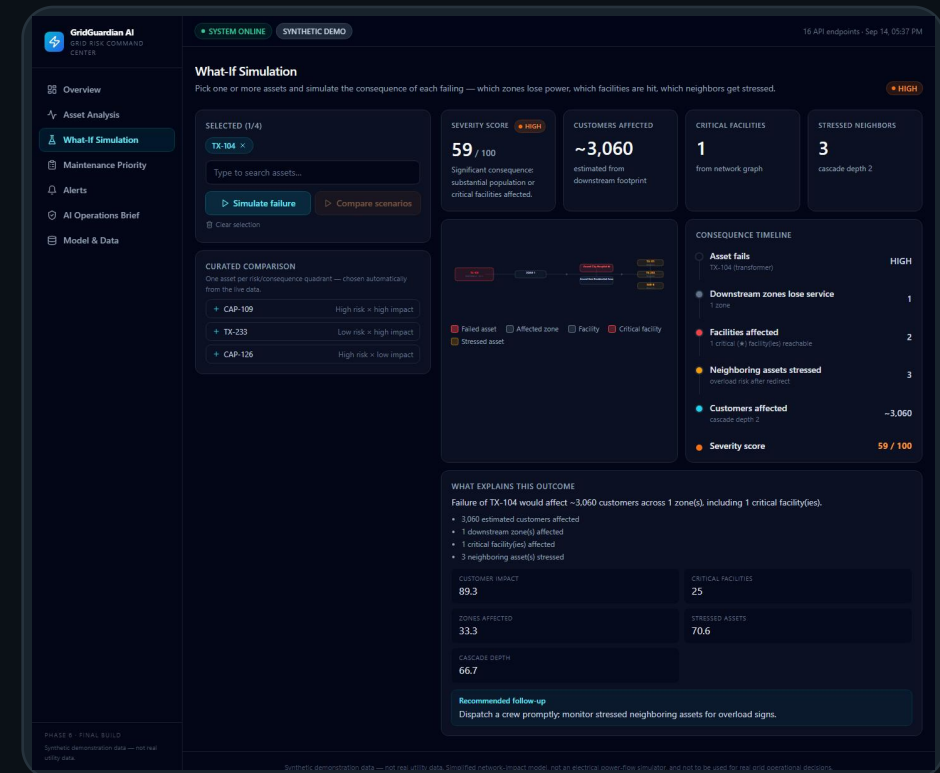
Risk-only thinking sends the crew to the wrong transformer.

TX-104 risks failing at 98.32% but affects ~3,060 customers. TX-233 has 0.52% risk — yet its failure would hit 5,377 customers and a hospital.

A decision copilot that chains four honest systems



- Failure probability (0–100) and risk category for every asset — not a black box.
- Graph-based impact: zones, critical facilities, customers, stressed neighbors.
- What-if on any asset, or compare four scenarios side by side.
- Transparent priority score and crew allocation that shows the uncovered gap.
- AI-ops brief generated strictly from structured backend facts.



A strict, testable pipeline end to end

DATA

800 scored assets · sensors · network graph

ML FAILURE PREDICTION

GradientBoostingClassifier · 21 features

GRID IMPACT ASSESSMENT

networkx blast-radius walk

WHAT-IF SIMULATION

cascade: zones · facilities · customers

PRIORITY ENGINE

risk × impact × urgency + crew allocation

STRUCTURED RESULTS

JSON contract enforced by API tests

AI OPERATIONS BRIEF

anthropic narration — key-gated

OPERATIONS DASHBOARD

React · Vite · Tailwind · Recharts

BACKEND — FastAPI + uvicorn

- Thin routers; logic only in tested services.
- Strict error contract: 400 / 404 / 422 / 500.
- pydantic-validated request/response models.

ML & SIMULATION — deterministic

- scikit-learn + pandas; seed 42, reproducible.
- networkx 3.3 graph cascade for what-if.
- metrics.json + refs to split (640 / 160).

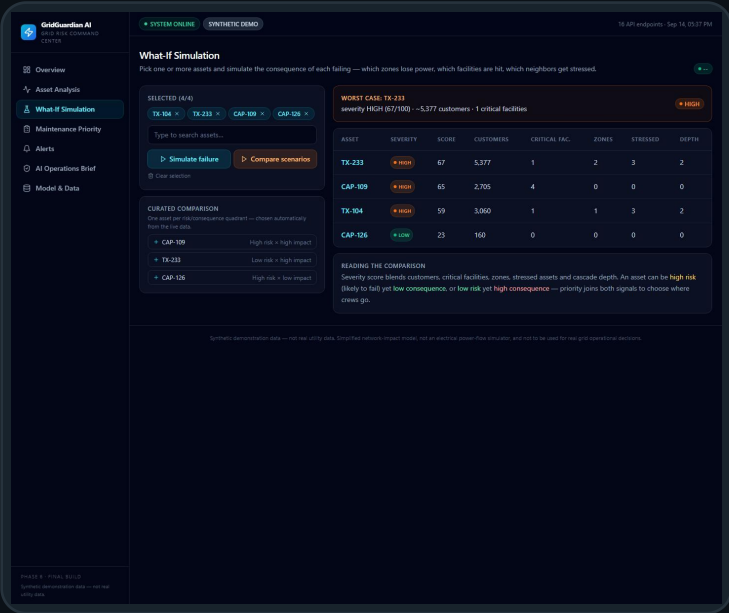
FRONTEND — React dashboard

- 8 operator pages with risk views.
- Recharts visuals, dark operator theme.
- Deep-linkable what-if: /simulate?ids=...

Failure risk is not the same as operational impact

Metric	TX-104 — transformer	TX-233 — transformer
Predicted failure risk	98.32% HIGH	0.52% LOW
Grid impact score	70.97	82.98
Maintenance priority	71.29 HIGH	0.44 LOW
Simulated severity (if failed)	HIGH — 58.62	HIGH — 66.97
Customers affected (if failed)	~3,060	~5,377
Critical facilities downstream	HOSP-1	HOSP-4 (ZONE-7/8)

Risk says “watch TX-104”. Consequence says “protect TX-233”.



Live — four what-if scenarios compared side by side

Five layers, each with a named output

ML FAILURE PREDICTION — *how likely*

Gradient boosting probability per asset — the “when”.

GRID IMPACT ASSESSMENT — *how big*

Deterministic blast-radius across the network — the “blast”.

WHAT-IF SIMULATION — *what breaks next*

Cascade walk: zones, facilities, customers, severity score.

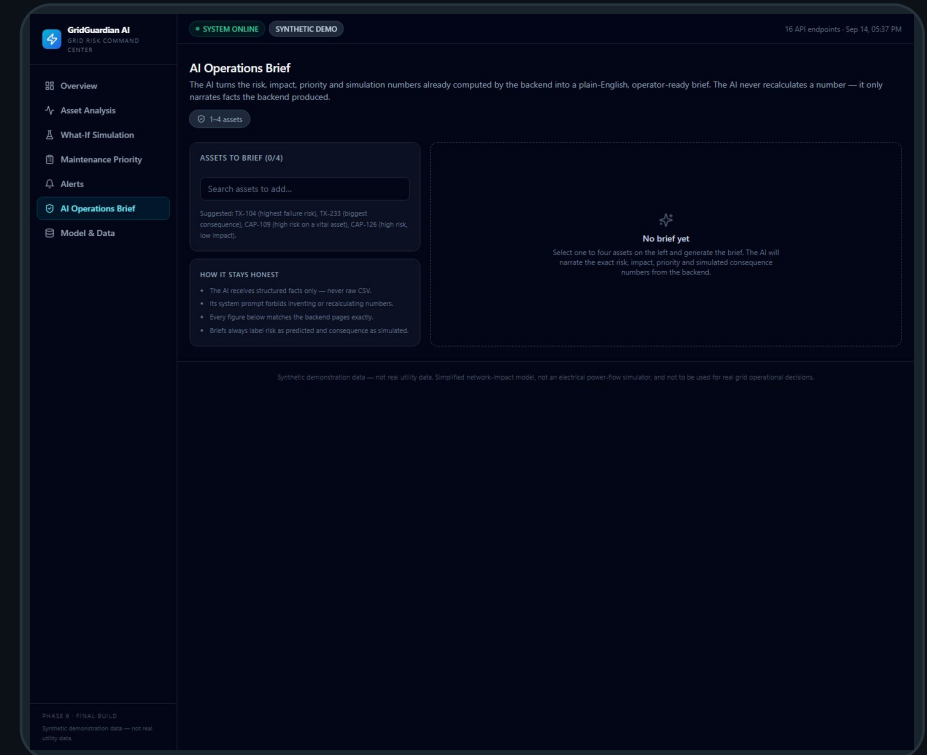
MAINTENANCE PRIORITY — *what to fix first*

Risk × impact × urgency score + crew allocation — the “call”.

GENERATIVE AI BRIEF — *so anyone can act*

LLM narration built from structured facts only.

Generative AI is a narration / decision-support layer — it never calculates ML risk or impact.



Live — AI operations brief with the facts used

Measured on the scoring engine's own output

800

grid assets monitored

0.8672

ROC-AUC on the 160-asset
holdout

60

open alerts — 8 critical

80

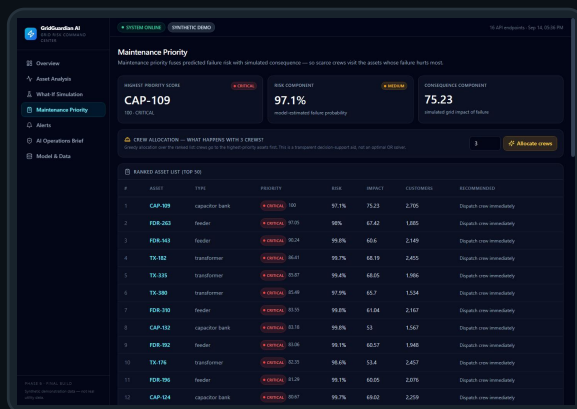
critical / high-priority assets

RISK DISTRIBUTION ACROSS ALL ASSETS



● CRITICAL 65 ● HIGH 96 ● MEDIUM 160 ● LOW 479

MAINTENANCE PRIORITY ● CRITICAL 21 ● HIGH 59 ● MEDIUM 105 ● LOW 615



HONEST LIMITATIONS

- Synthetic, seeded demonstration data — labeled in-app, not real utility data.
- Cascade simulation is a deterministic graph walk — not a power-flow solver.
- Without an API key the app still works and says what to do.

Supports 155,751 customers behind critical / high-priority assets.

TEAM

VOLTVISION · AI Track

Aum Vaja — Team Lead

Vrunda Kotadiya — Member

Tarang Prajapati — Member

Riza Bhatt — Member

WHY IT MATTERS

“The grid doesn’t give operators more time — it gives them more data. GridGuardian turns that data into a decision, before the grid forces one.”

Predict the risk. Simulate the failure. Prioritize the crew. Respond — before it happens.

Synthetic demonstration data — not real utility data. Hackathon prototype.